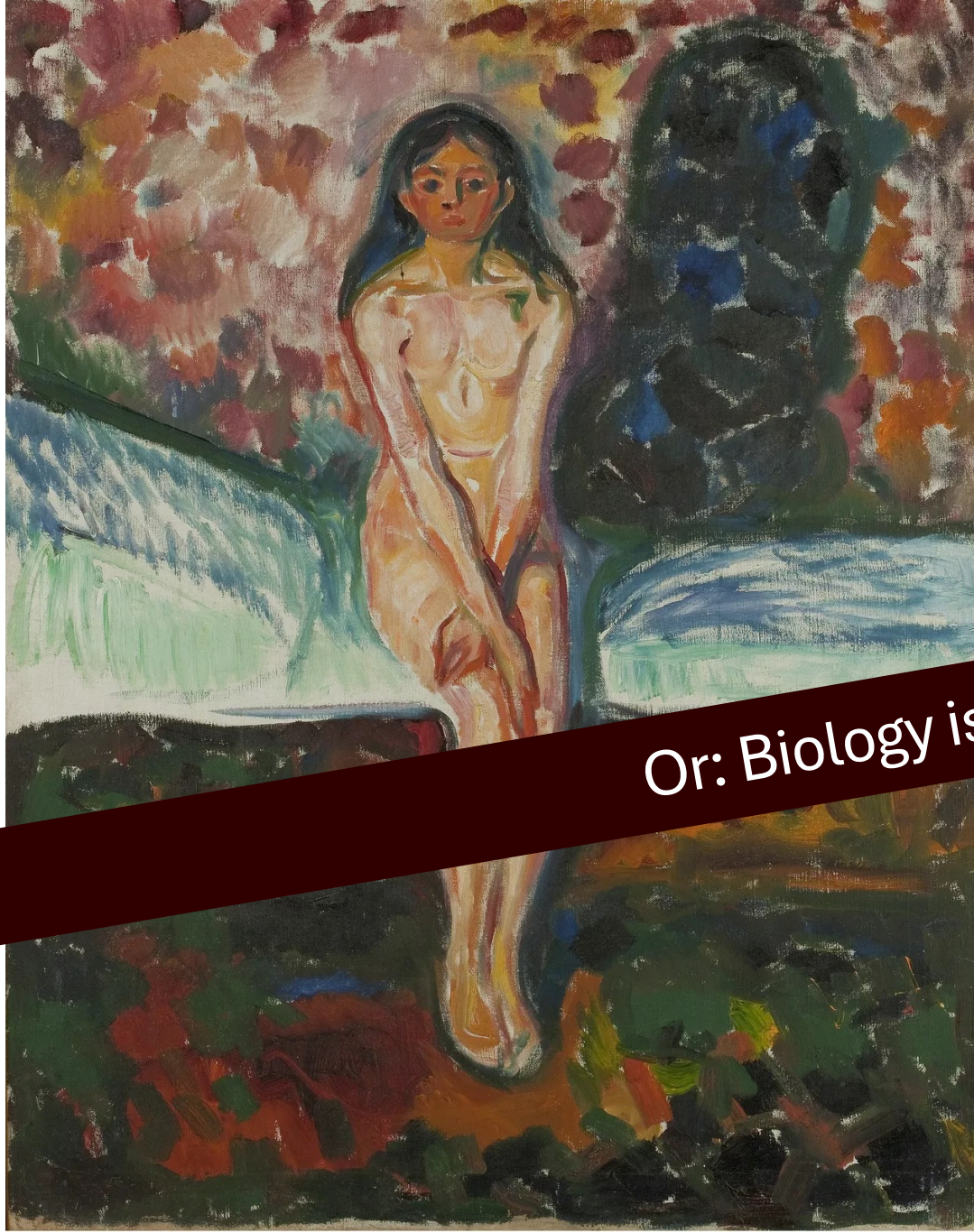


**CN:**

- Detailed descriptions and images of internal and external genitalia during development
- Brief discussion of energy restriction (anorexia nervosa & weight restriction in sports)

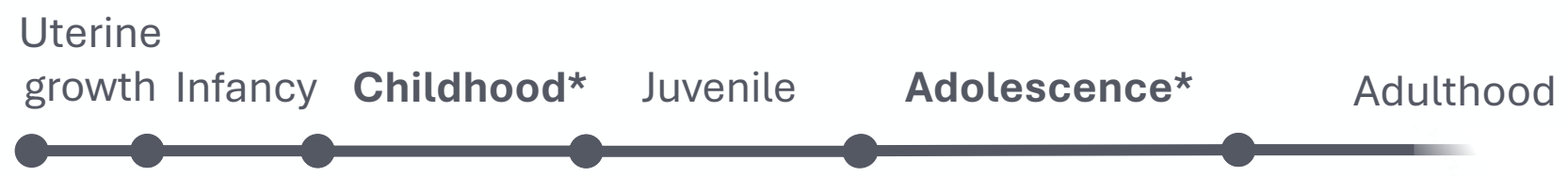
Or: Biology is the study of continua

Reproduction I

Dr. Laura van Holstein

lavanho@emory.edu

Room 215



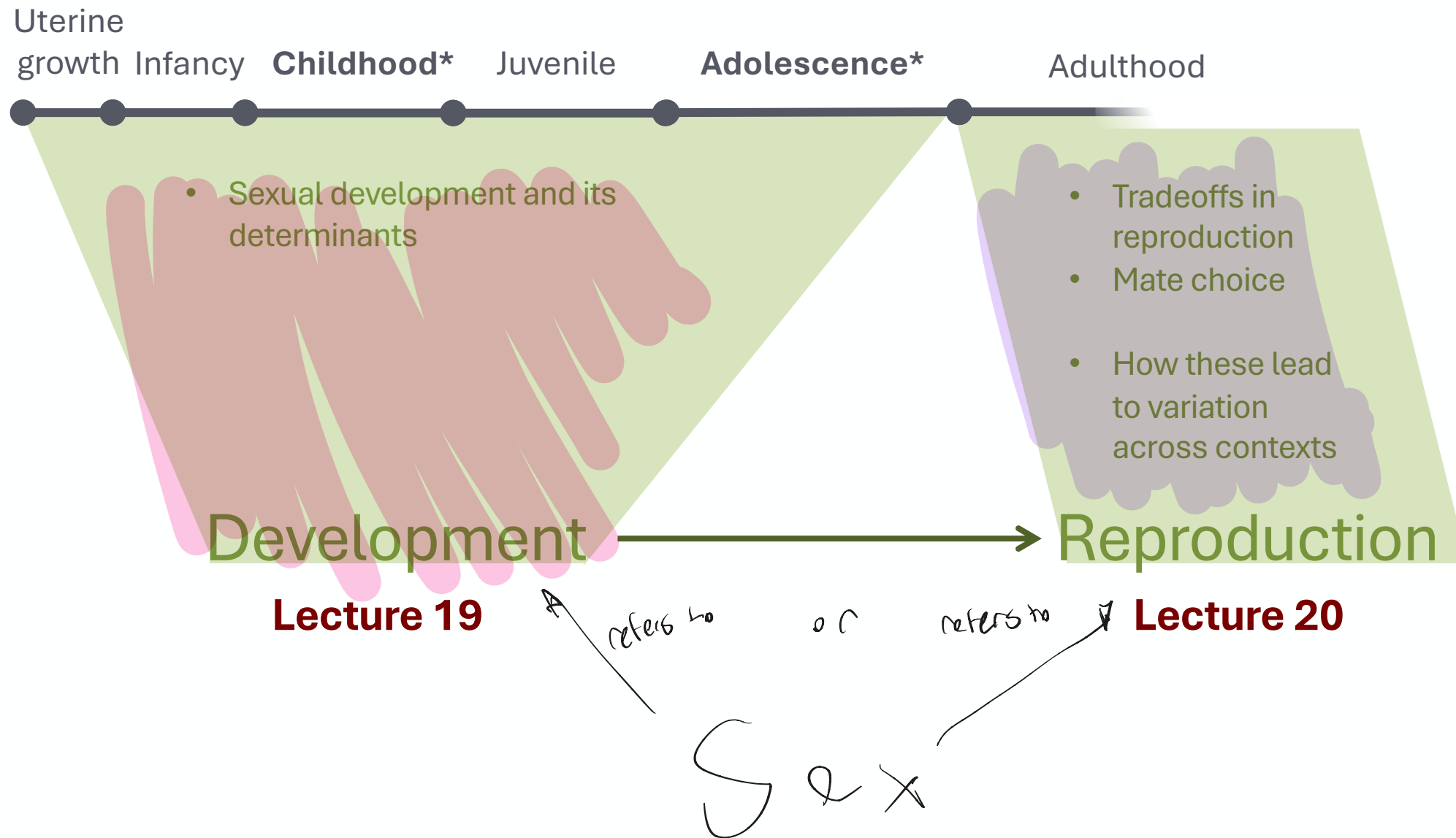
Development → Reproduction

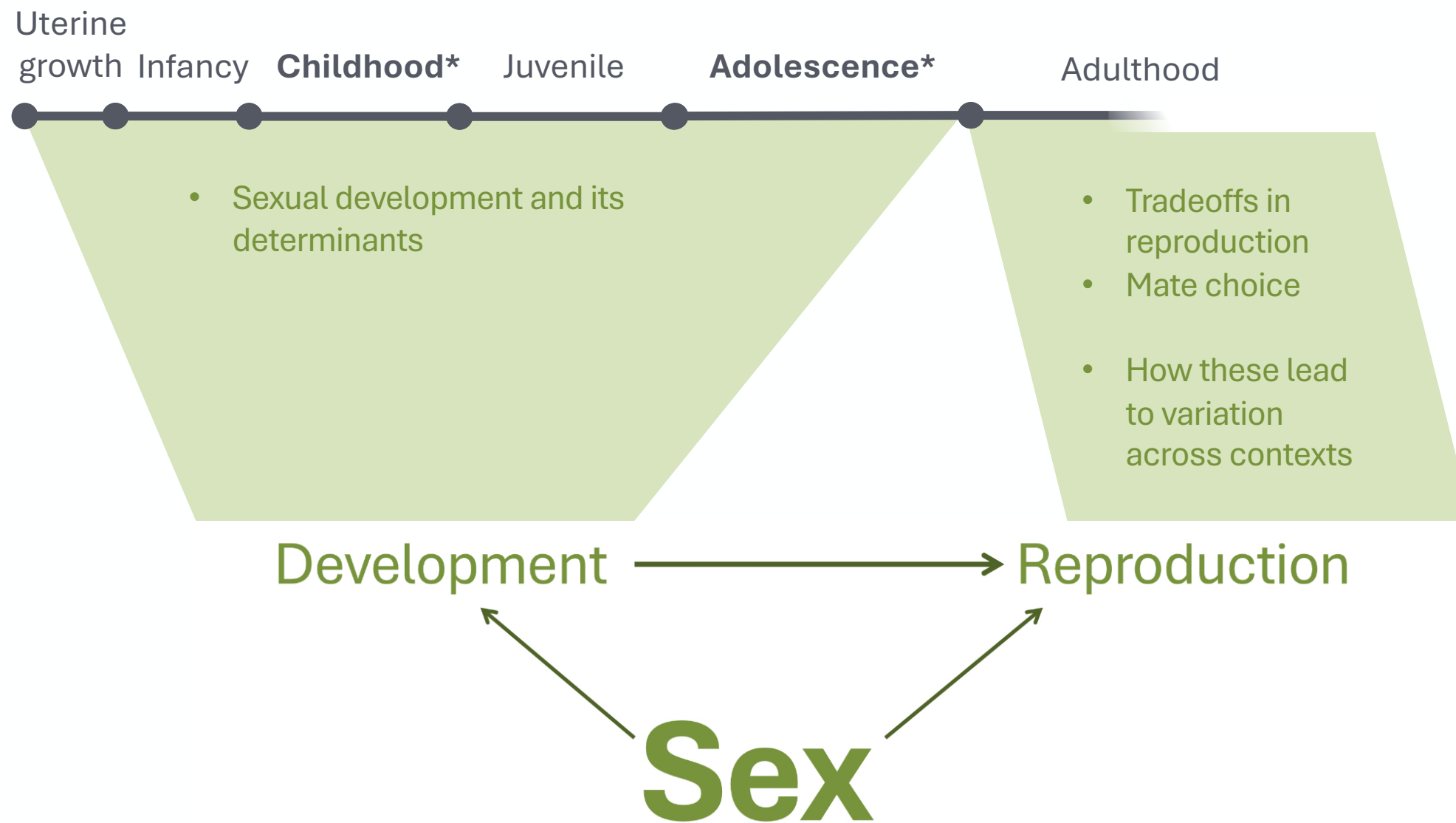
Adult forms differentiated based on gametes + ontogenetic variation about how these forms developed

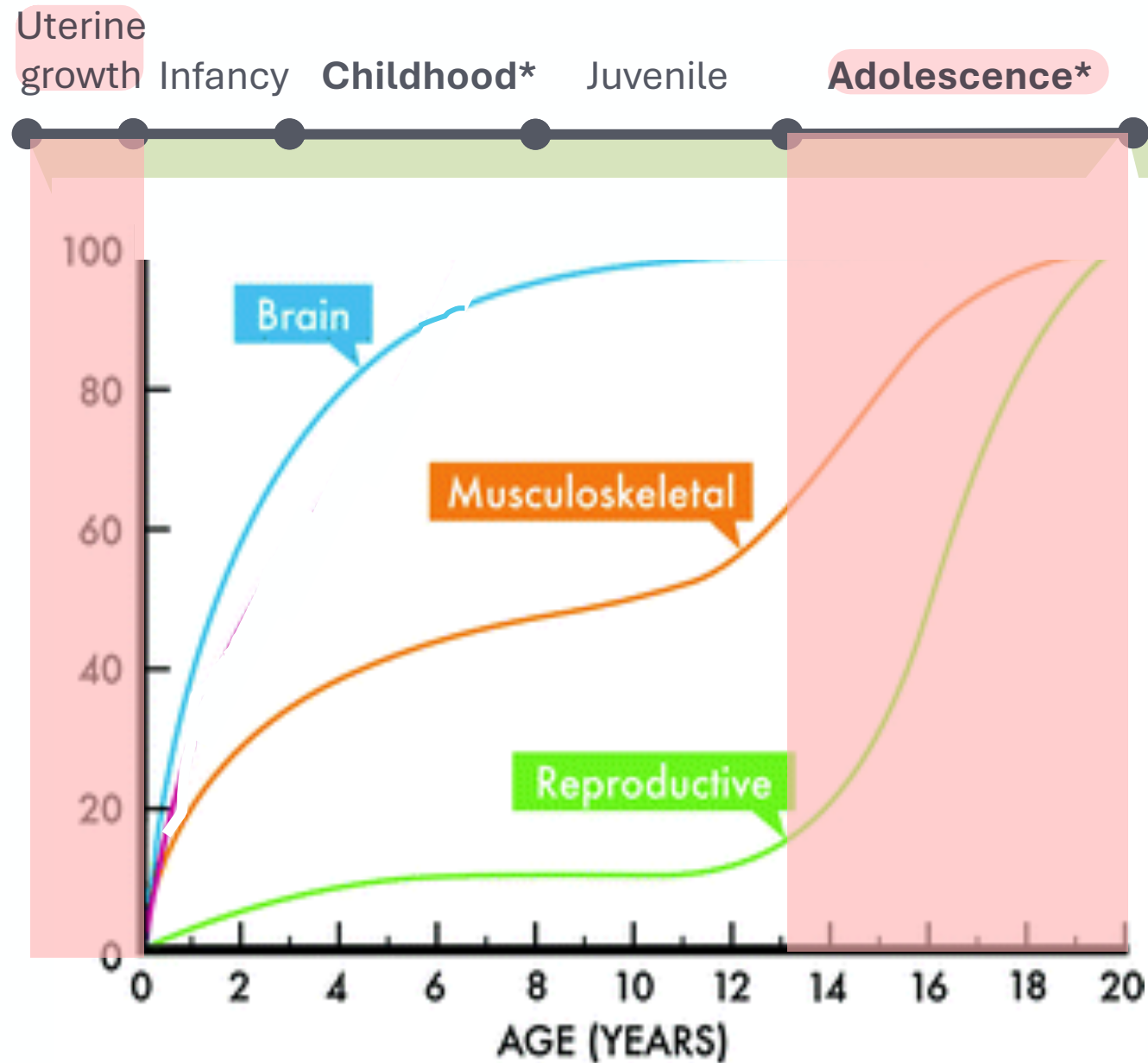
Sex

Inter course → process of reproduction → Fundamental point of life

mating, joining, sex, taking care of offspring



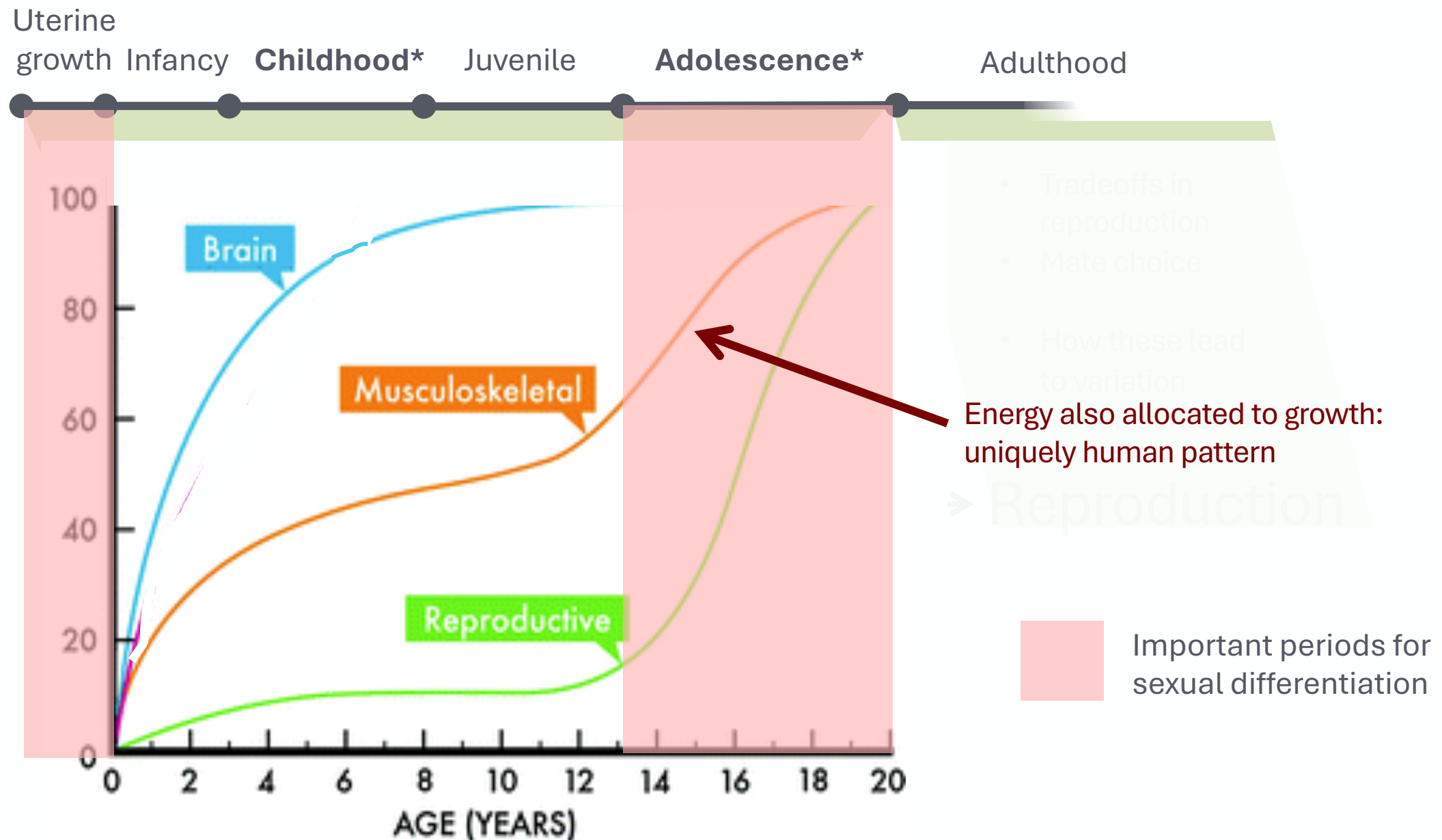




- Tradeoffs in reproduction
- Mate choice
- How these lead to variation across contexts

➤ Reproduction

Important periods for sexual differentiation



General
primate
pattern -
interest
in male/
female
designation
↳ Inherited
from
LCA

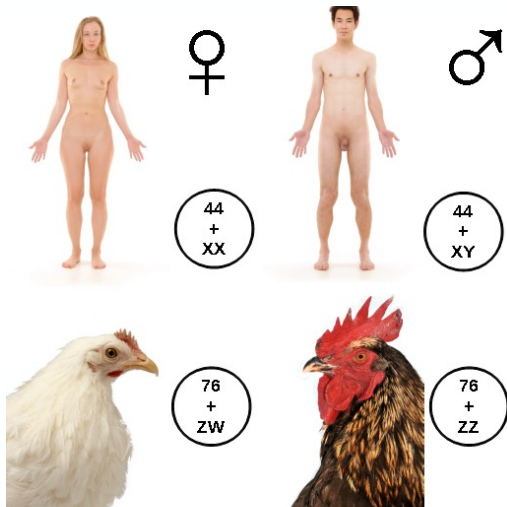


Context

Sex

Adolescent growth

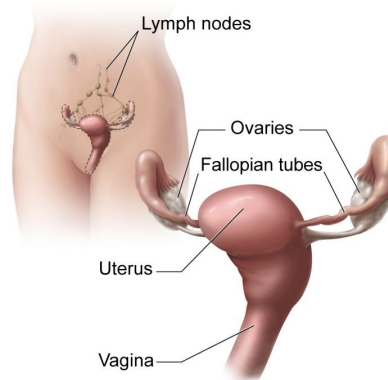
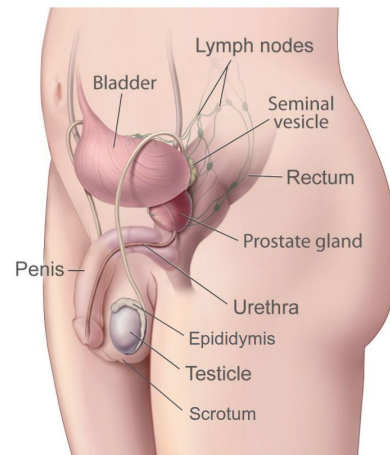
Chromosomes



Gonads

Endocrine

Genitalia



Brains

Secondary sexual characteristics

- Larynx
- Muscles
- Breasts
- Skeletal size
- ...

↳ Phenotypic sex

Chromosomes

XY

Gonads

Testicles

Endocrine

T>E

Genitalia

Born with
penis

Brains

Secondary sexual
characteristics

A sex difference in the human brain and its relation to transsexuality

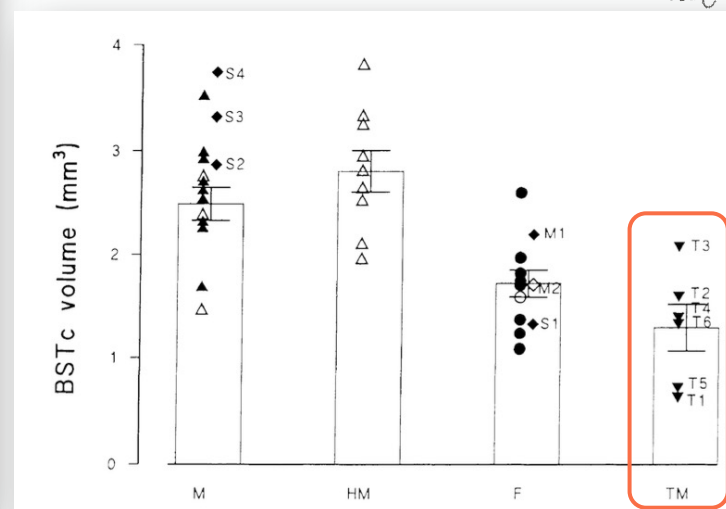
Jiang-Ning Zhou*, Michel A. Hofman*,
Louis J. G. Gooren† & Dick F. Swaab*‡

* Graduate School Neurosciences Amsterdam,
Netherlands Institute for Brain Research, Meibergdreef 33,
1105 AZ Amsterdam ZO, The Netherlands

† Department of Endocrinology, Free University Hospital,
1007 MB Amsterdam, The Netherlands

TRANSSEXUALS have the strong feeling, often from childhood onwards, of having been born the wrong sex. The possible psychogenic or biological aetiology of transsexuality has been the subject of debate for many years^{1,2}. Here we show that the volume of the central subdivision of the bed nucleus of the stria terminalis (BSTc), a brain area that is essential for sexual behaviour^{3,4}, is larger in men than in women. A female-sized BSTc was found in male-to-female transsexuals. The size of the BSTc was not influenced by sex hormones in adulthood and was independent of sexual orientation. Our study is the first to show a female brain structure in genetically male transsexuals and supports the hypothesis that gender identity develops as a result of an interaction between the developing brain and sex hormones^{5,6}.

Important to know they found a region of the brain where males & females differ in size even in homosexuality, but trans women have a size in this region most similar to females - and hormone levels to not make a difference here



Trans women break "expected" cascade -
Chromosomes gonads, hormones, genitalia
are "male" but brain & secondary
chars are not male

Trans women
have brain size
regions most
similar to females

Chromosomes

Gonads

Endocrine

Genitalia

Brains

Secondary sexual
characteristics

Two key points:

1. There are strong correlations between these, but $R^2 \neq 1$!
2. There is no absolute binary at any of these levels

Chromosomes

XX

XY

X (Turner syndrome)

XXY (Klinefelter syndrome)

XXX (Trisomy X)

XYY

Gonads

Endocrine

Genitalia

Brains

Secondary sexual
characteristics

Chromosomes

Gonads

Testes

Ovaries

Testes + Ovaries
(46XY DSD)

Endocrine

Genitalia

Brains

Secondary sexual
characteristics

Chromosomes

Gonads

Endocrine

Genitalia

Brains

Secondary sexual characteristics

Guevedoces (Dominican Republic)



Growing a Penis @ Puberty

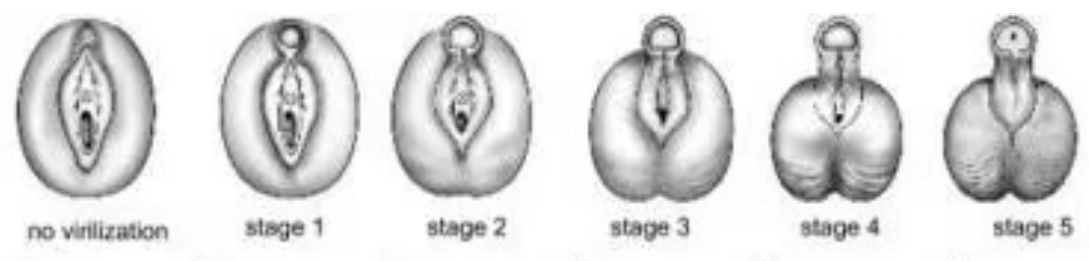
Penis
Vagina

Continuum (e.g.,
5a reductase
deficiency – XY)

Androgen
insensitivity

Genitals derived from the same structures so there can be ambiguous genital development

no receptiveness to testosterone



Chromosomes

Gonads

Endocrine

Genitalia

Non-binary

Brains

More common than IQs >140

Secondary sexual characteristics



The usual figure is 1.7%

having deviation from binary @
a given level is more common than IQ > 140

Chromosomes

↓
What is
invariant is
that the sex
Chrs are Xs
and Ys

Gonads

Endocrine

Genitalia

Brains

Secondary sexual characteristics

Two key points:

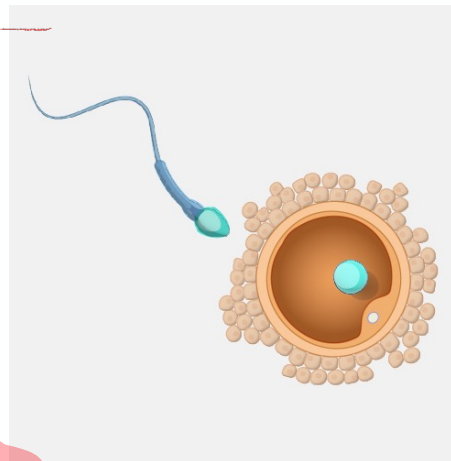
1. There are strong correlations between these, but $R^2 \neq 1$!
2. There is no absolute binary at any of these levels

apart from

- the existence of the sex chromosomes (ish)
- gametes

→ variance
in combination

gametes
explicitly
binary



No spergs or speggs!

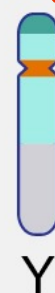
Chromosomes

- 8th largest chromosome
- Highly conserved

↓
Mammal
X chrs similar
across species



X



Y

- **SRY gene:** acts as switch; inhibits female development, and initiates male development

ads

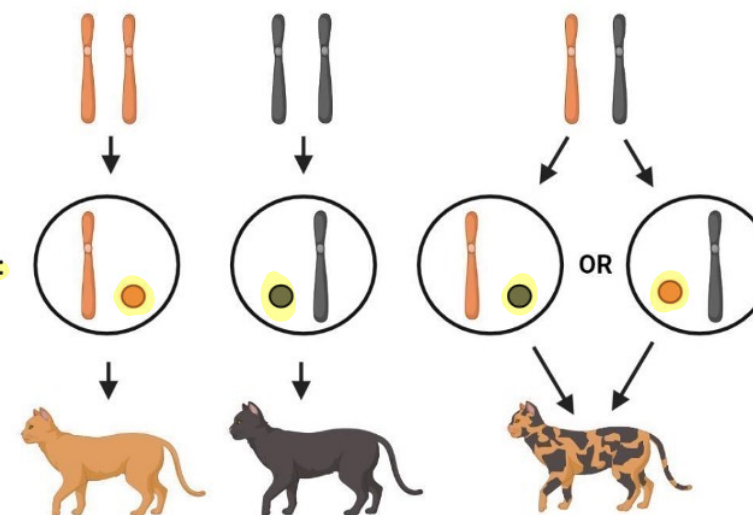
Endocrine

Ge

XX genotype:

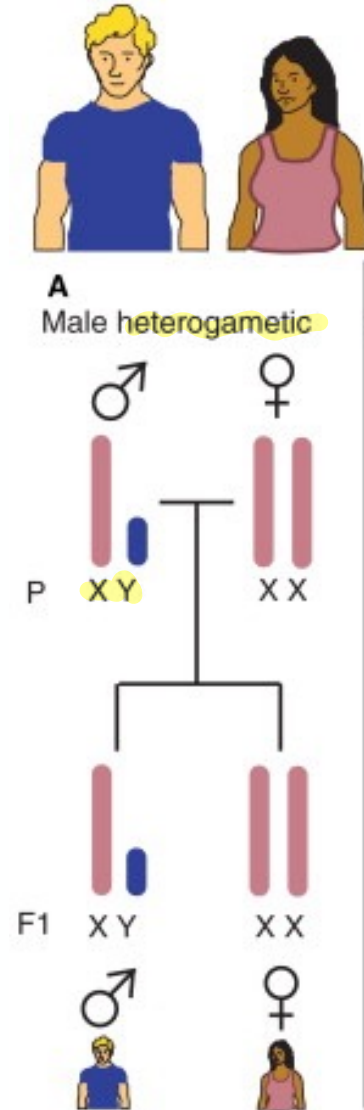
Chromosomes
after X inactivation:

Fur color:



patches from inactive of diff color?

Chromosomes



ads

Endocrine

Genitalia

Brains

Secondary sexual
characteristics

When is it set?

Context

Sex

Adolescent growth

Chromosomes

Gonads

Endocrine

Genitalia

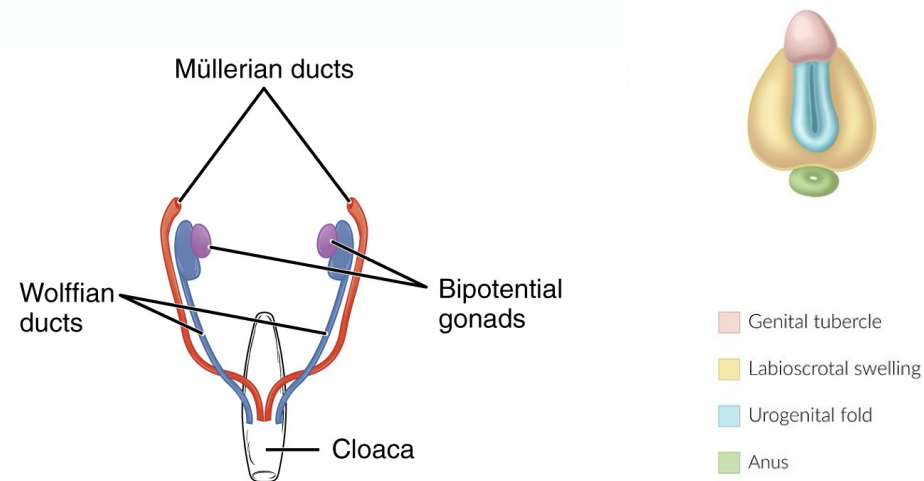
Brains

**Secondary sexual
characteristics**

Fertilization

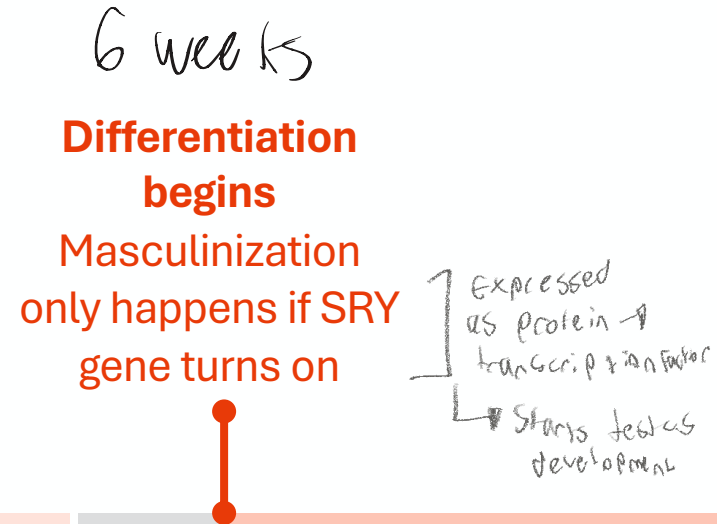
1st trimester

2nd trimester

**Indifferent stage: 0-6 weeks**

- **Gonads:** Genital ridges form
- **Internal genitalia:** 2 ducts form
- **External genitalia:** Basic bipotential structures form

Fertilization**1st trimester****2nd trimester**



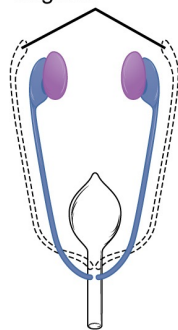
Fertilization

1st trimester2nd trimester

Masculinization

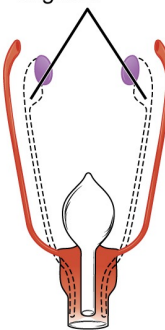
Feminization

Müllerian ducts
degrade

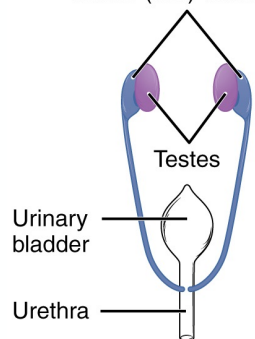


*Bbf
external
genitals
not
directly
correlating*

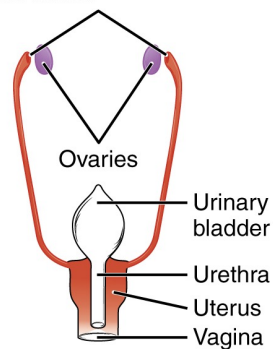
Wolffian ducts
degrade



Wolffian ducts
become epididymis and
ductus (vas) deferens

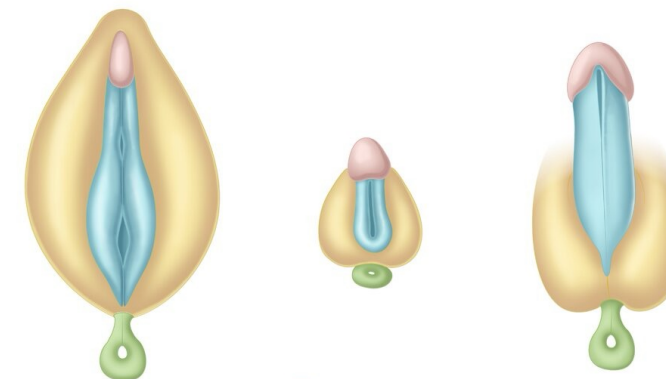


Müllerian ducts
become uterine tubes
and uterus



Differentiation

- **Gonads:** Testes develop in males, and **secrete T. T acts on bipotential structures** → *External genitals*
- **Internal genitalia:** Degradation of Müllerian ducts in males, Wolffian ducts in females
- **External genitalia:** e.g., T ensures that genital tubercle becomes glans penis



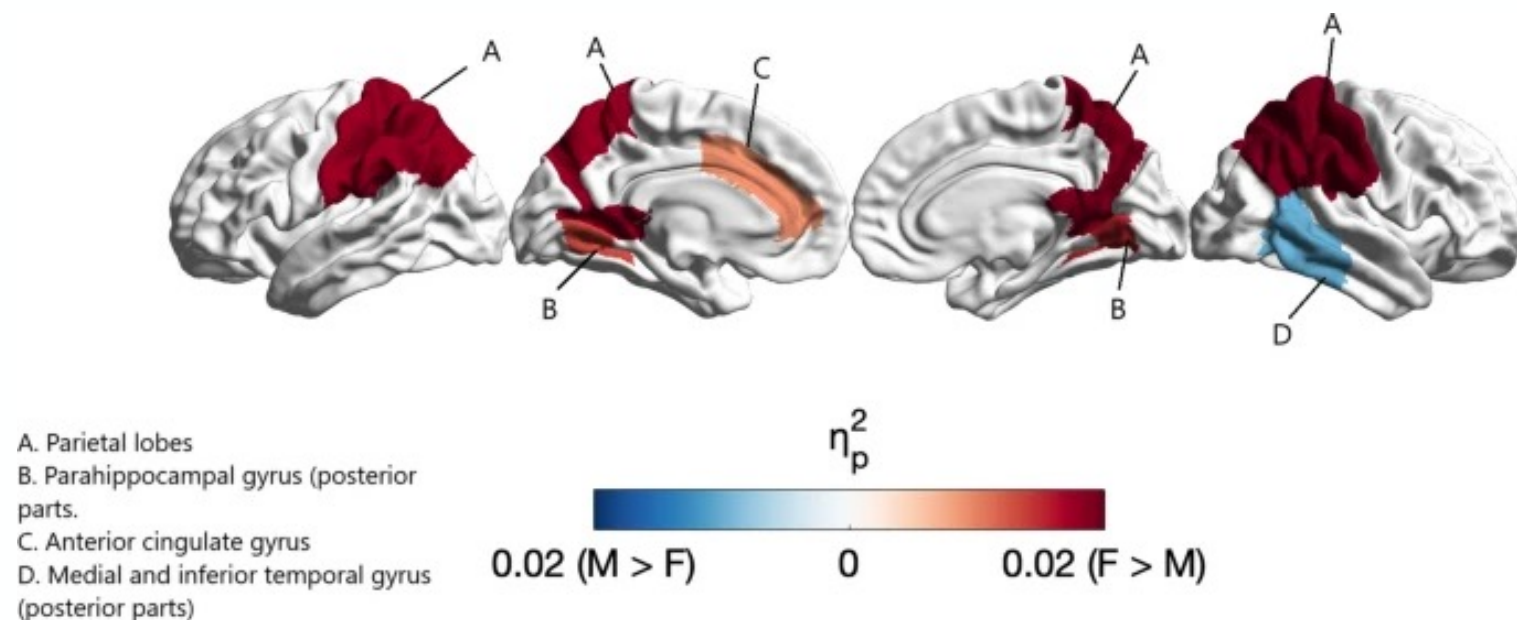
Genital tubercle
Labioscrotal swelling
Urogenital fold
Anus

**~Week 14:
differentiation
complete**

Fertilization

1st trimester

2nd trimester

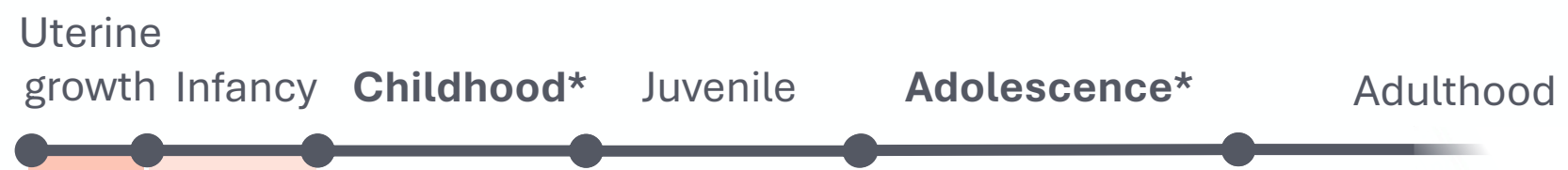
**Take-home message:**

Several sex differences that have previously been observed later in development are present from birth, emphasising the key role that **prenatal factors** play in initiating sex differences in the brain.

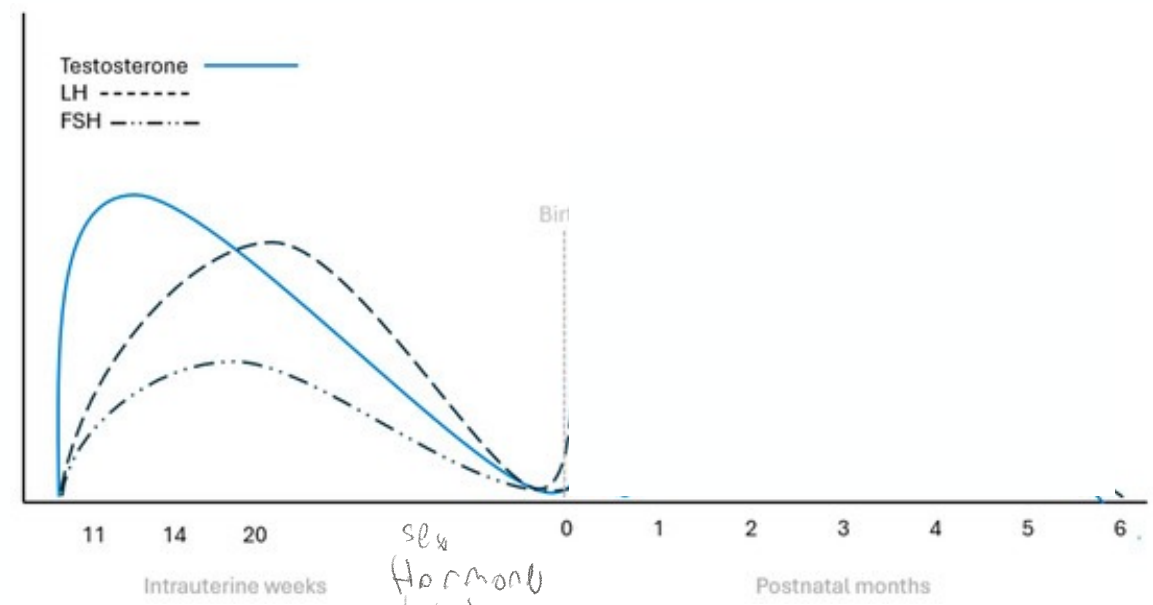
Brain differentiation

- Brains are sexually dimorphic at birth – here are some examples:
 - M larger brains, but F more gray matter
 - F bigger than male: corpus callosum, bilateral parietal lobes
 - M bigger than female: right medial temporal gyri

Fertilization**1st trimester****2nd trimester**



- Sexual differentiation at **very high** rates
- Growth of reproductive organs at **very high** rates
- Genetic & endocrine control



sex hormone levels drop once uterine development is done but they pick back up again shortly after birth

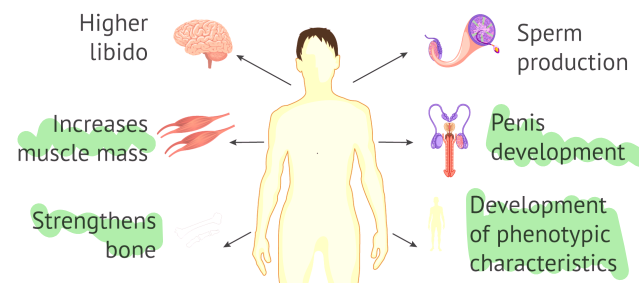
Uterine growth Infancy **Childhood*** Juvenile **Adolescence*** Adulthood



Minipuberty

- First week of life: hypothalamus secretes GnRH
 - GnRH stimulates **gonadotropins** (LH and FSH)
 - 6-9 months: LH declines in girls
 - 6-9 months: both decline in boys
- 1-3 months: **T** at pubertal levels in boys
- 6-12 months: **Estradiol** high in girls

Testosterone:



WHAT?!

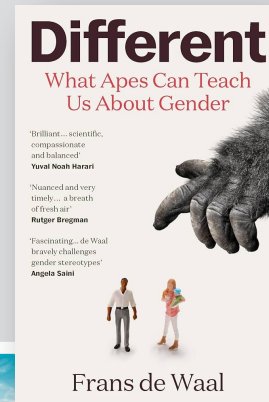
Current thinking suggests that it is an **essential imprinting period** for different body functions. *Telling the body what to do when it does grow*

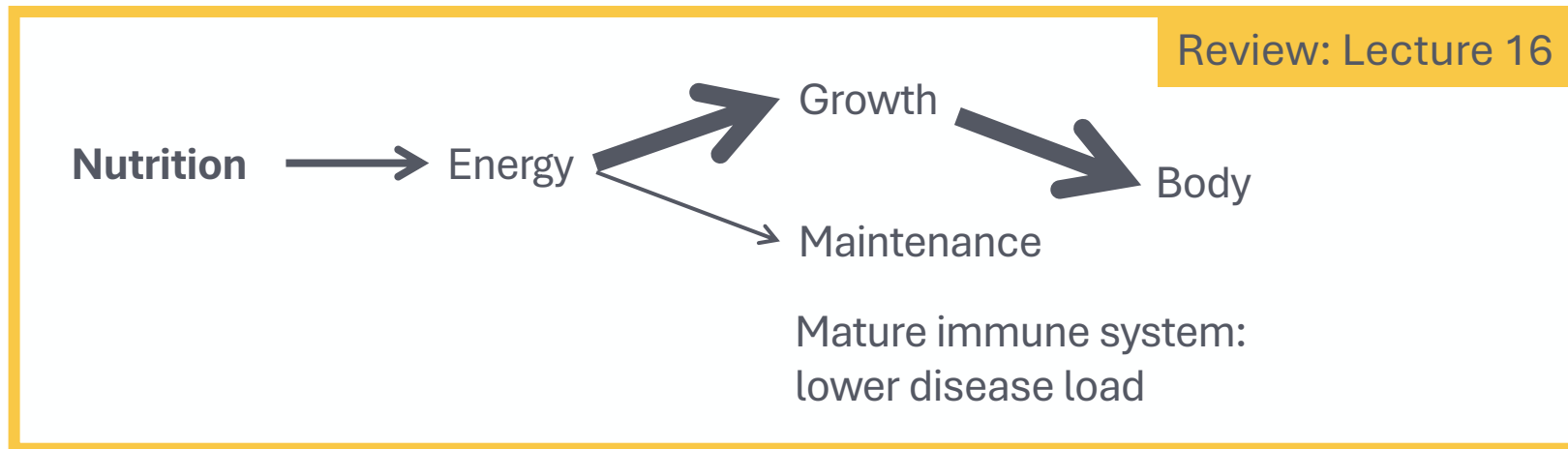
- Important role in genital organ development (e.g., T influences penile growth, the number of Sertoli cells, and spermatogenesis).
- Influences the infant's body composition (e.g., T affects BMI, body weight, and growth velocity)
- Affects cognitive functions; T has an impact on language organization in the infant brain and estradiol affects laryngeal sound production and baby babbling.

Becker, Marianne, and Volker Hesse. "Minipuberty: why does it happen?." *Hormone Research in Paediatrics* 93.2 (2020): 76-84.

Uterine growth Infancy **Childhood*** Juvenile **Adolescence*** Adulthood

Not much biology, but...
socialisation (gender!)







Growth starts following distinct trajectories for males and females:

Uterine

growth Infancy

Childhood*

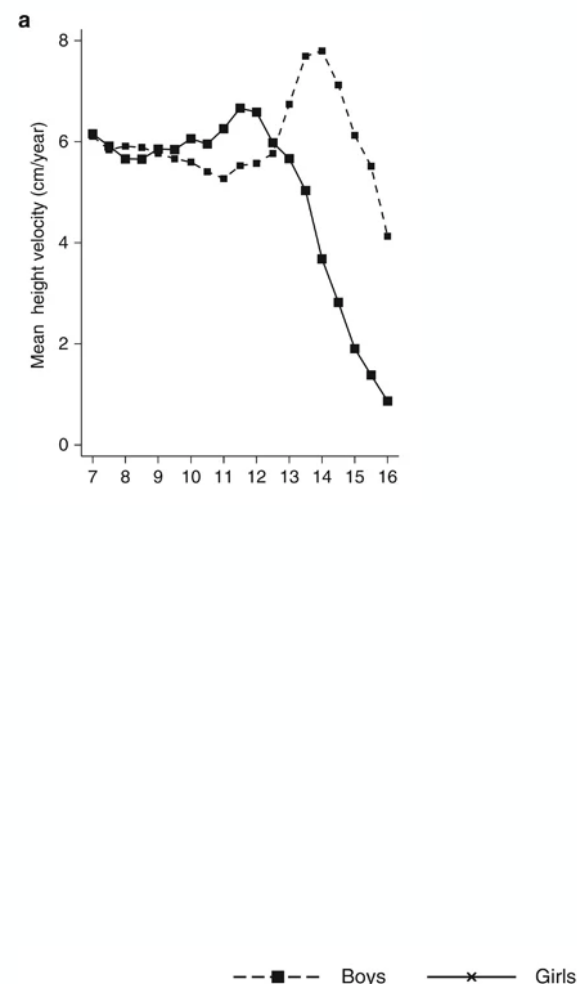
Juvenile

Adolescence*

Before adolescence
indistinct growth rate trajectories most

Growth starts following distinct trajectories for males and females:

- Later and higher peak height velocity in boys → end up being taller



Uterine

growth Infancy

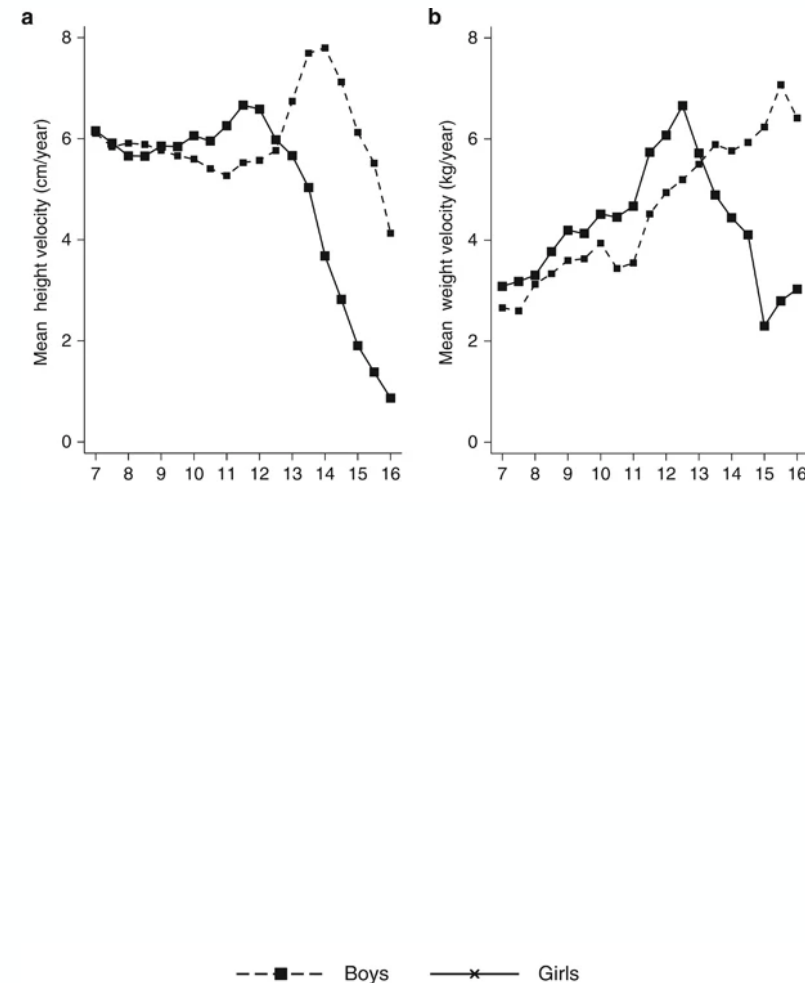
Childhood*

Juvenile

Adolescence*

Growth starts following distinct trajectories for males and females:

- Later and higher peak height velocity in boys
- Later peak weight velocity in boys



Uterine

growth Infancy

Childhood*

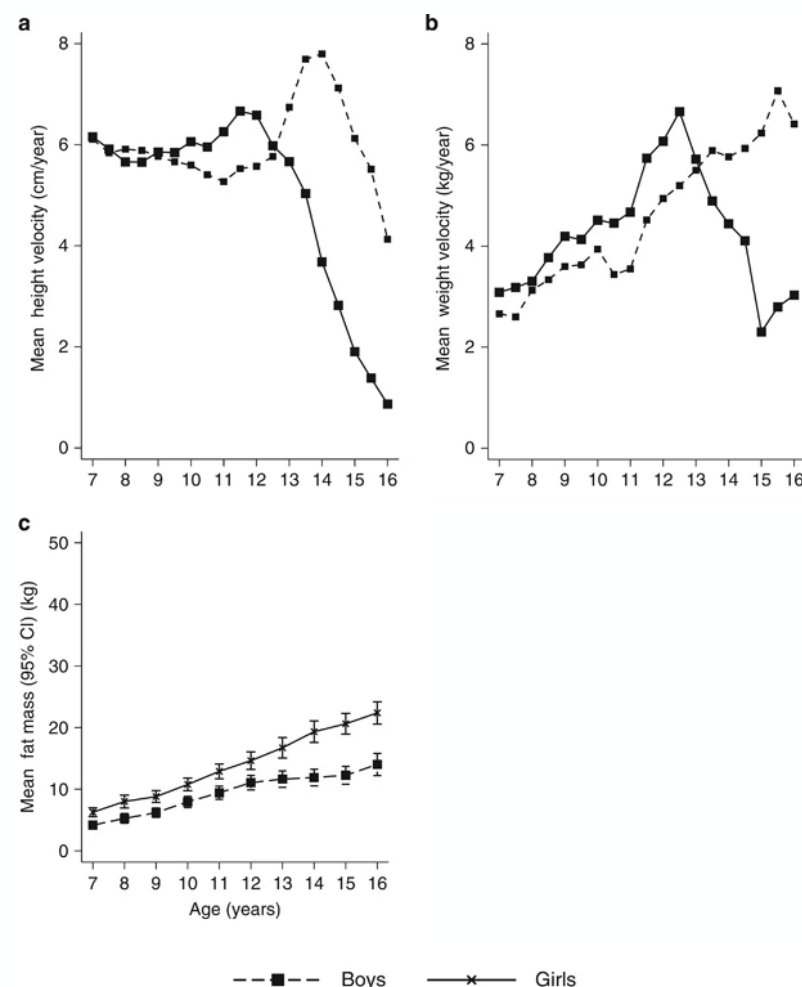
Juvenile

Adolescence*



Growth starts following distinct trajectories for males and females:

- Later and higher peak height velocity in boys
- Later peak weight velocity in boys
- Increasing difference in relative fat mass: more in girls



Uterine

growth Infancy

Childhood*

Juvenile

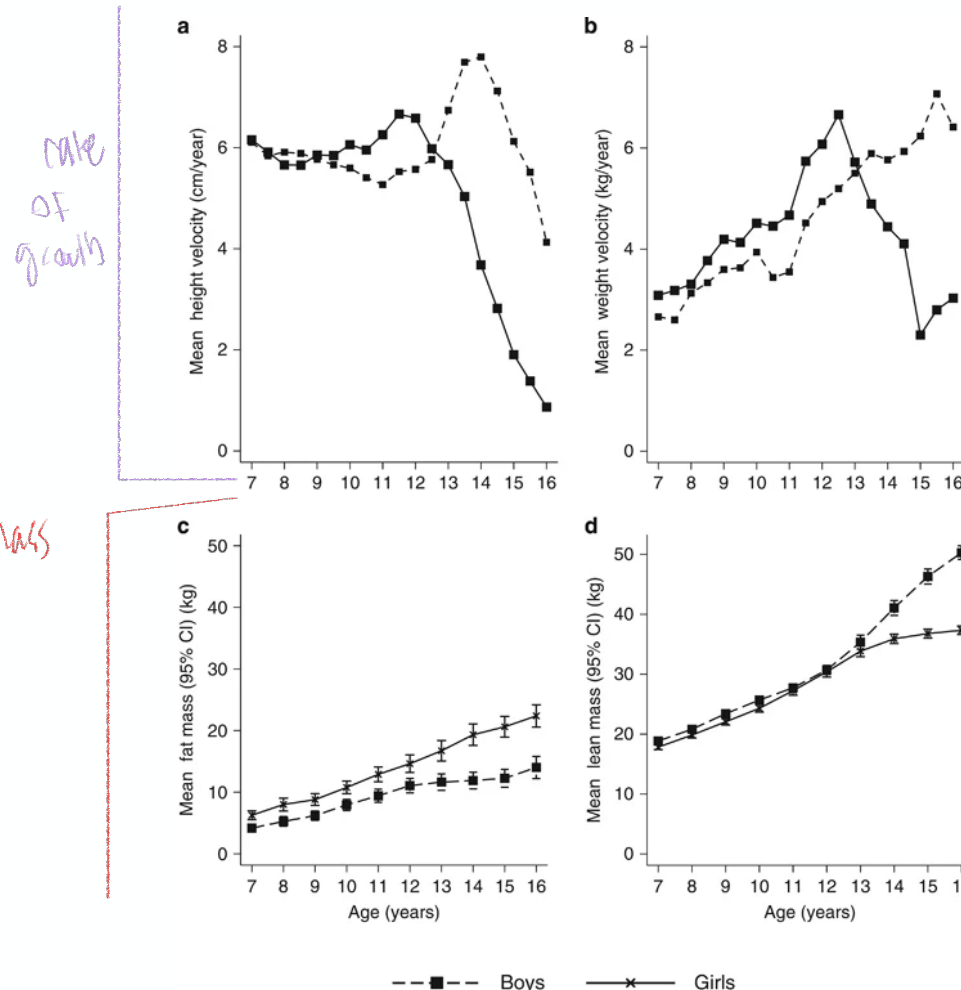
Adolescence*

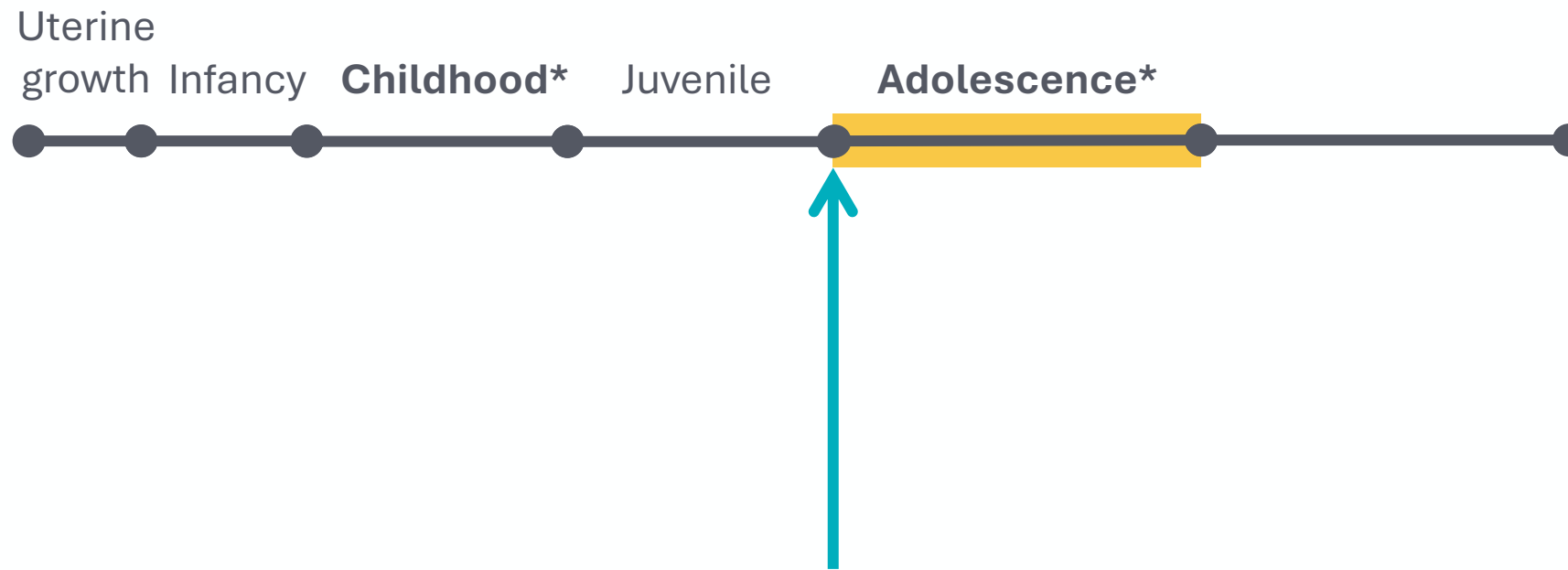


Growth starts following distinct trajectories for males and females:

- Later and higher peak height velocity in boys
- Later peak weight velocity in boys
- Increasing difference in relative fat mass: more in girls
- Increasing difference in muscle mass: more in boys

+ development of
Secondary sex charact.
crisis





What factors drive the
timing of this event?

↳ Switch from
Juvenile period
to adolescence



What is this event?



Reactivation gradually starts during juvenile period - pulses of secretion at night to reawaken axis

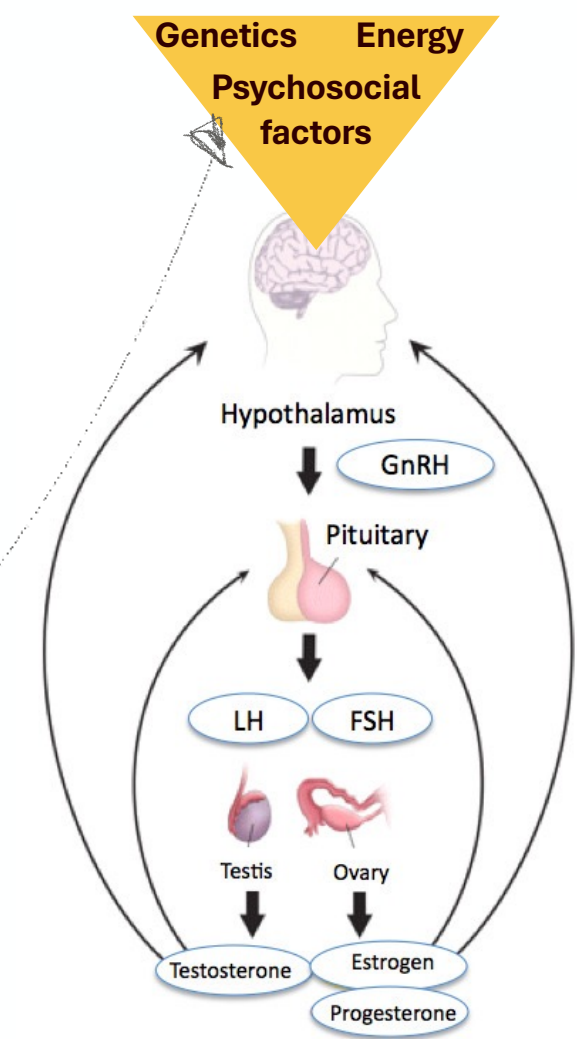
What is this event?

Reactivation of the hypothalamic-pituitary-gonadal (HPG) axis

- Timing determined by the 'hourglass of puberty' (Ellison et al., 2012)

What influences hypothalamus w.r.t this axis

was active in mini-puberty but dormant in childhood & juvenile period



Uterine

growth Infancy

Childhood*

Juvenile

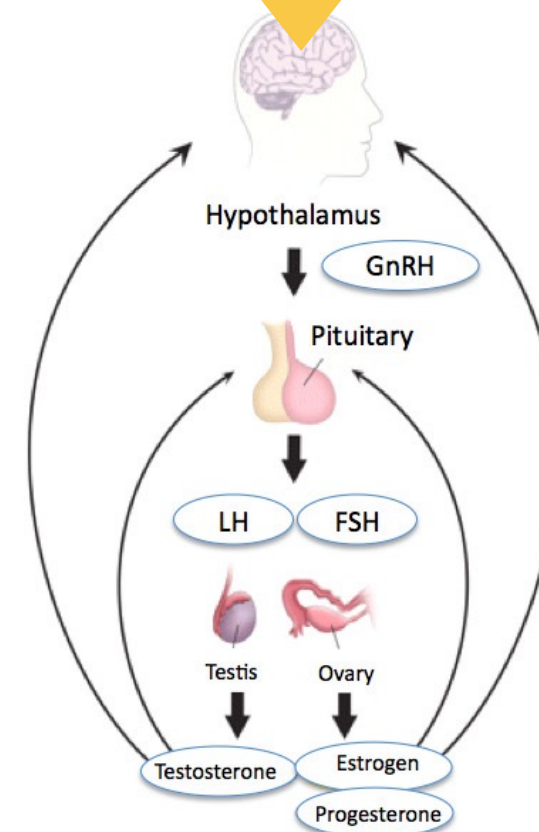
Adolescence*



What is this event?

- Actually measurable point is varied
 - Peak height velocity
 - Menarche
 - Spermarche
 - Tanner stage 2

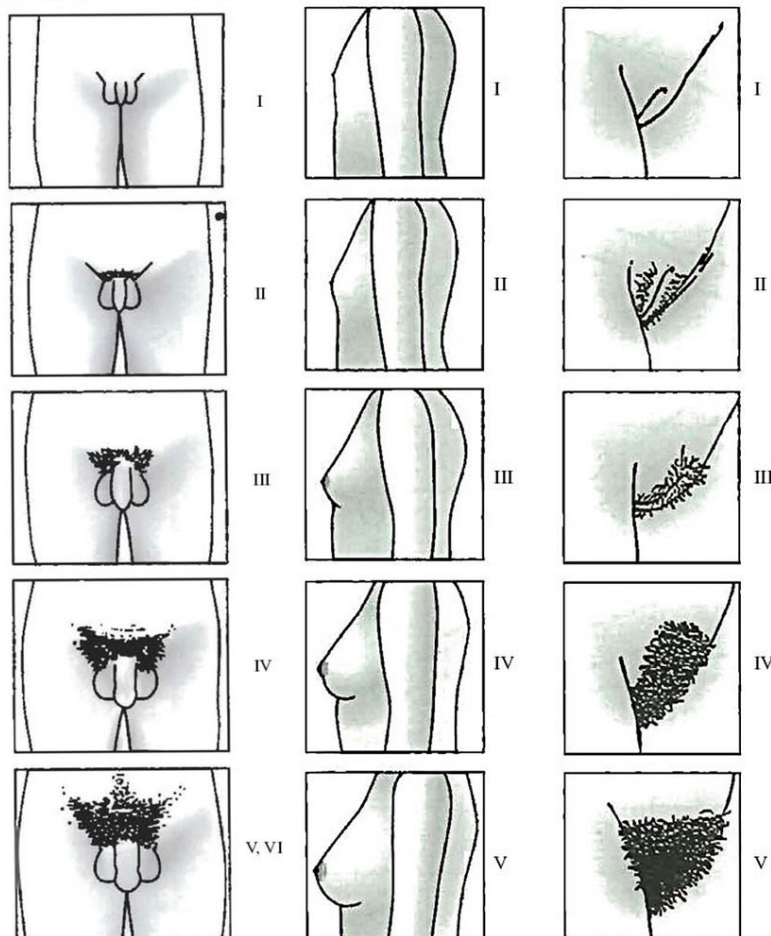
Genetics **Energy**
Psychosocial
factors

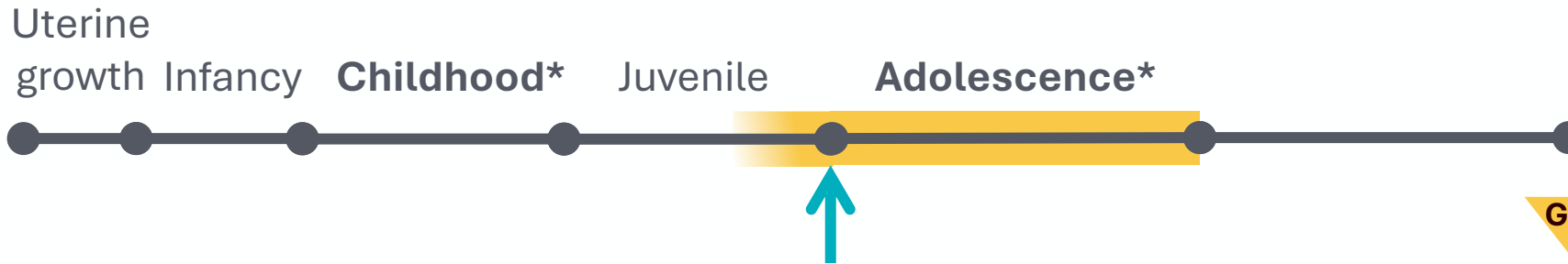


Genital stage and pubic hair stage

Breast stage

Pubic hair stage

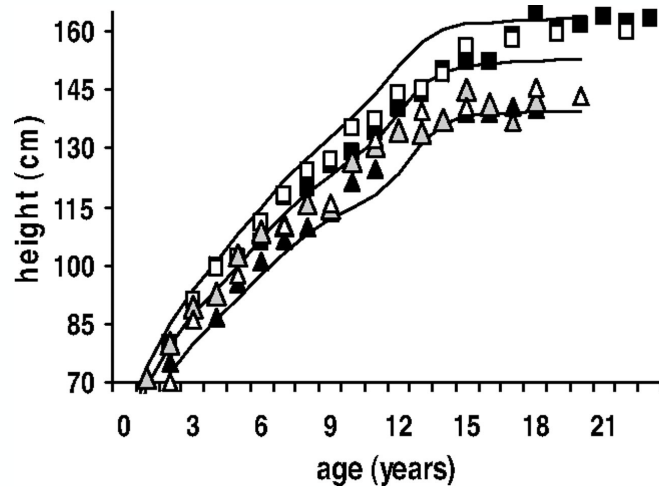




Role of genetics

- Studies done in affluent societies to control for environment affecting energy availability
 - Heritability between 0.5-0.9
 - Timing of growth spurt in Finnish boys: 0.9 (Silventoinen et al., 2008)
 - Timing of menarche in WEIRD societies: 0.5 (Towne et al., 2005)
- Relevant gene expression mediated by energy availability

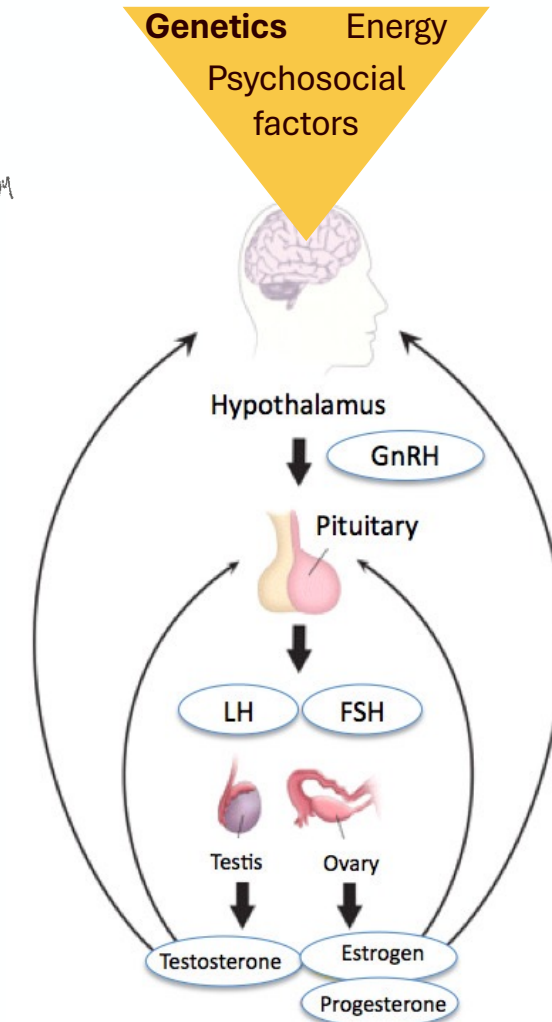
Review: Lectures 8 & 11



Research

Whole-genome sequence analyses of Western Central African Pygmy hunter-gatherers reveal a complex demographic history and identify candidate genes under positive natural selection

Ping-Hsun Hsieh,¹ Krishna R. Veeramah,^{2,3} Joseph Lachance,^{4,5} Sarah A. Tishkoff,⁴ Jeffrey D. Wall,⁶ Michael F. Hammer,^{1,2} and Ryan N. Gutenkunst^{1,7}



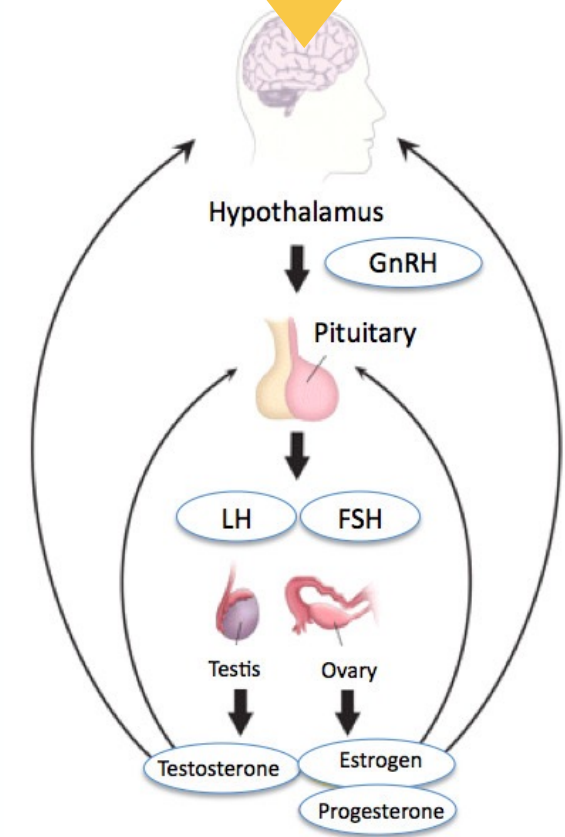


Role of energy availability

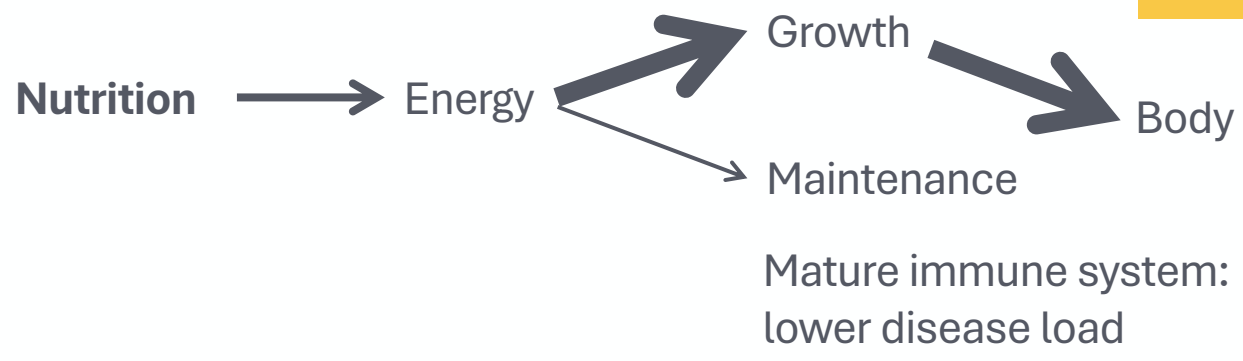
Infection drives energy availability here though!

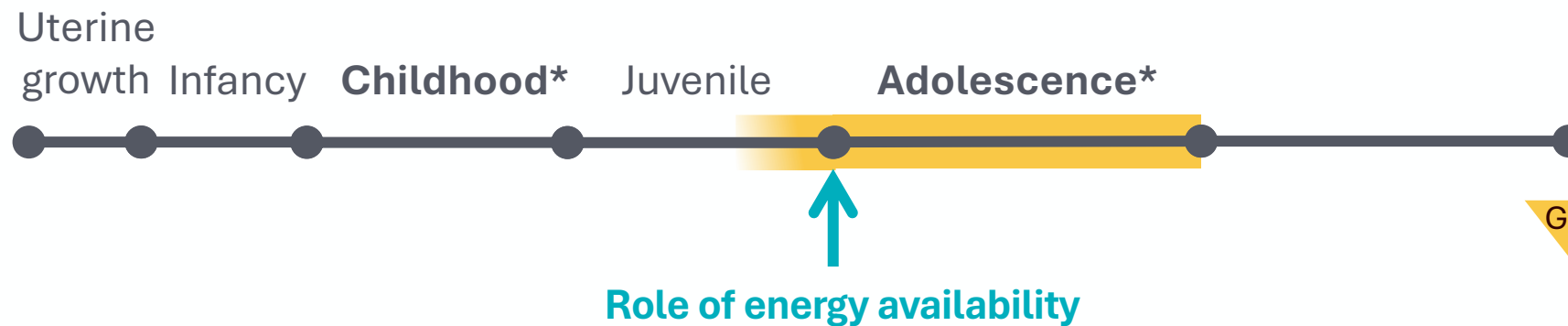
During 1st pulses to start endocrine complex

Genetics **Energy**
Psychosocial factors



Review: Lecture 16





Anorexia nervosa:

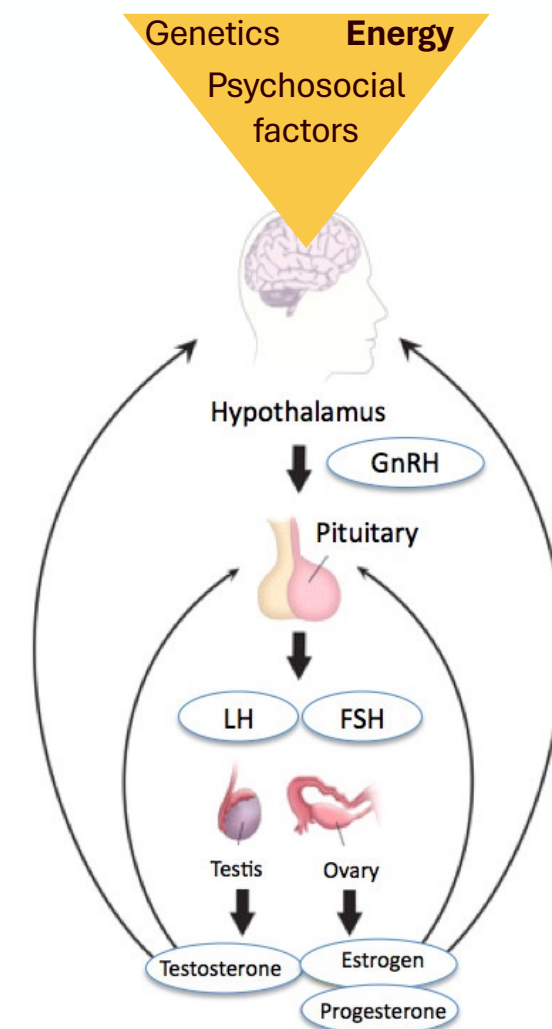
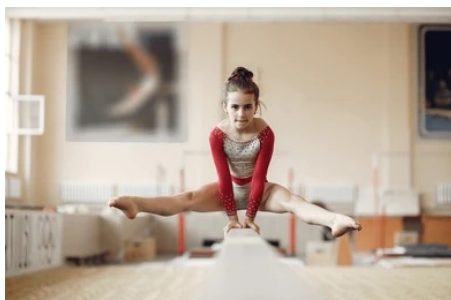
- Postpones menarche ^{one of a few potential markers of start of adolescence}
- Decreases pubertal growth overall (leading to shorter stature)
- Results in decreased ovarian function in adulthood

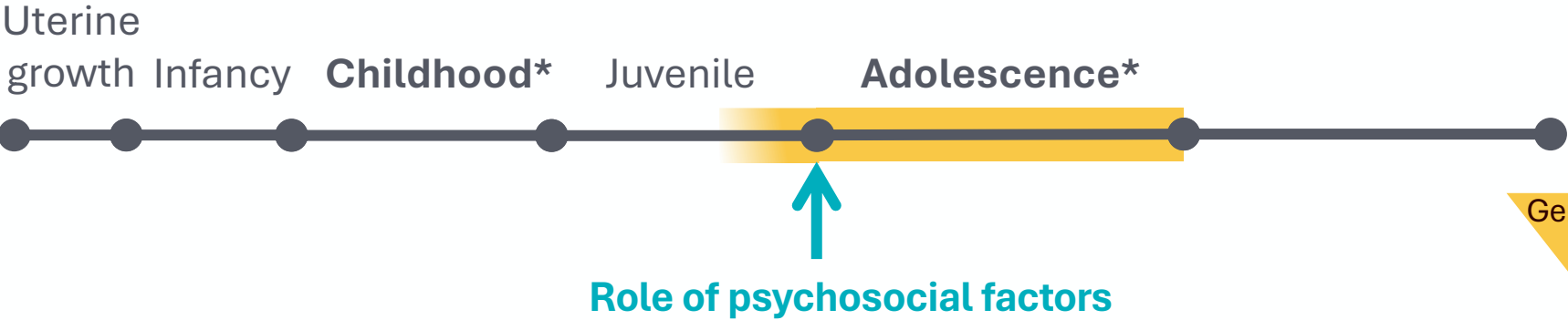
Swenne I, Thurfjell B. **2003**. Clinical onset and diagnosis of eating disorders in premenarcheal girls is preceded by inadequate weight gain and growth retardation. *Acta Paediatr* 92:1133–37

Athletes (gymnasts & wrestlers)

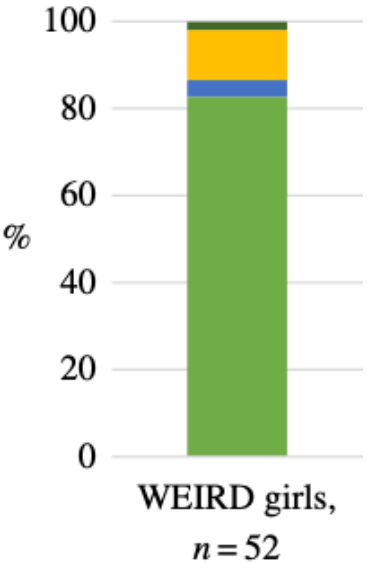
- Postpones menarche

Klentrou P, Plyley M. **2003**. Onset of puberty, menstrual frequency, and body fat in elite rhythmic gymnasts compared with normal controls. *Br. J. Sports Med.* 37:490–94



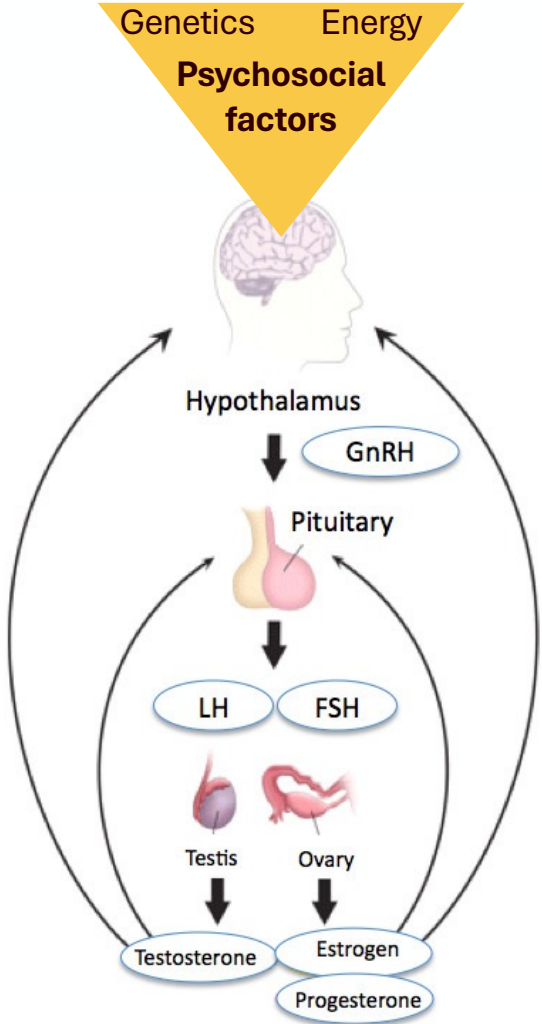


Percentage of studies that found accelerating, delaying, mixed or no associations between **father absence** and the timing of puberty



■ accelerates ■ delays ■ mixed ■ none/not stated

Why would this pathway evolve?





Role of psychosocial factors

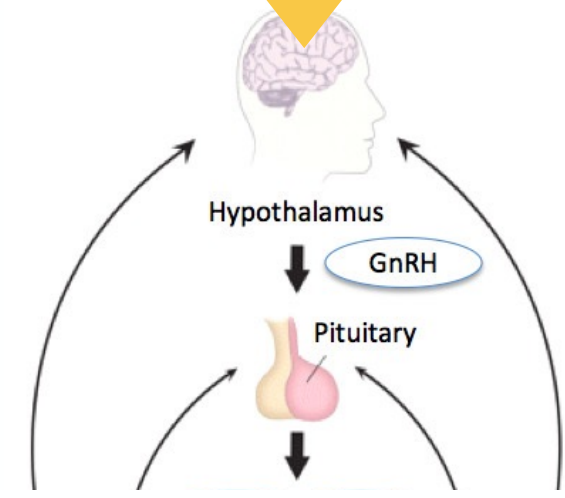
- Loss of a parent
- Divorce
- Poverty



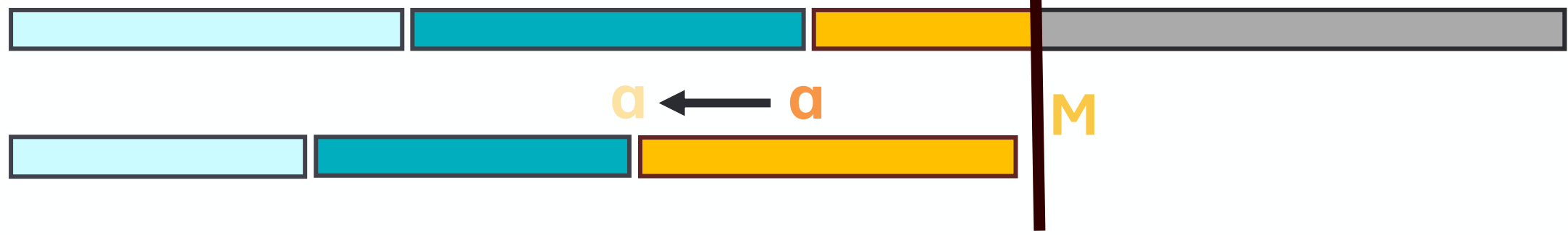
Communicate extrinsic mortality risk

↳ cue of unsafe environment

Genetics Energy
Psychosocial factors



Charnov's model:

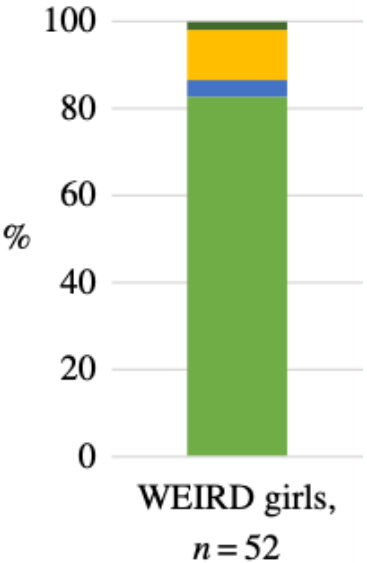




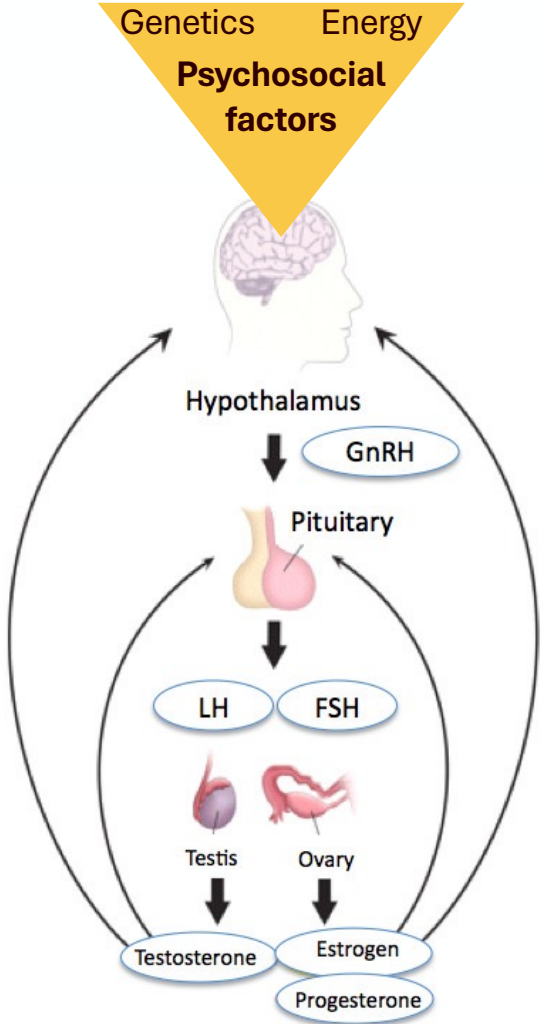
But....

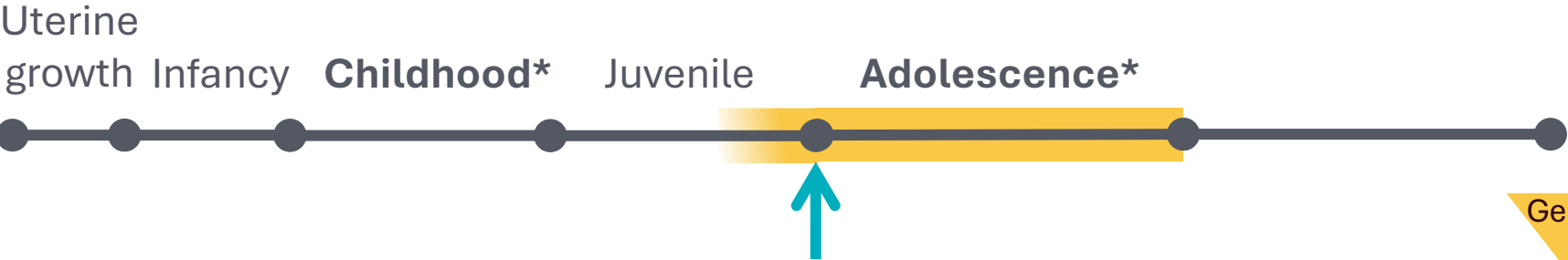
Role of psychosocial factors

Percentage of studies that found accelerating, delaying, mixed or no associations between **father absence** and the timing of puberty



■ accelerates ■ delays ■ mixed ■ none/not stated



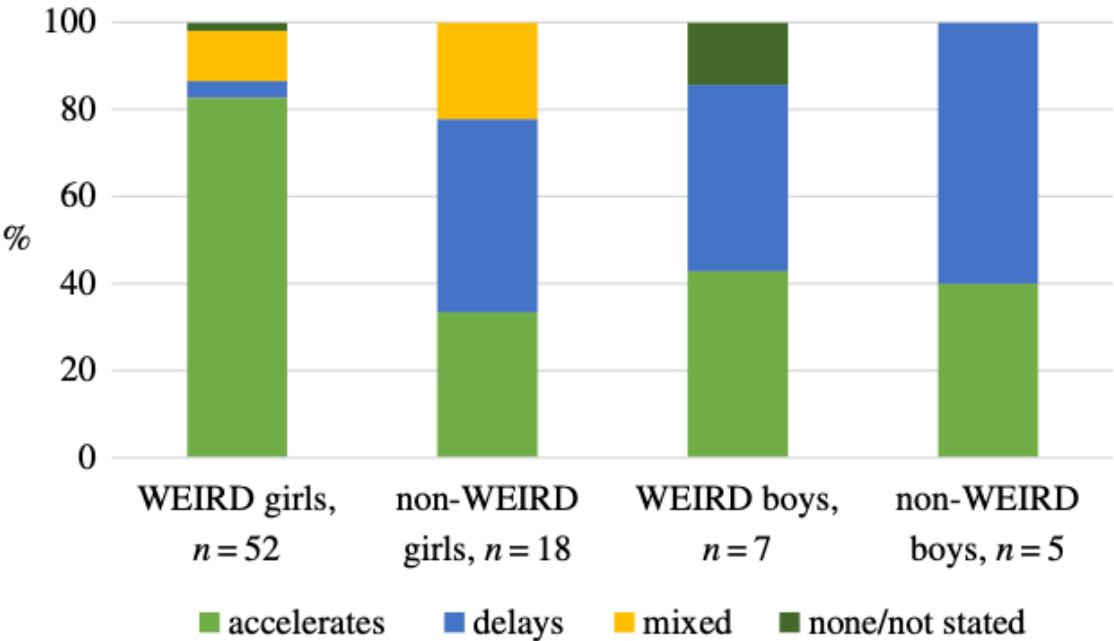


But....

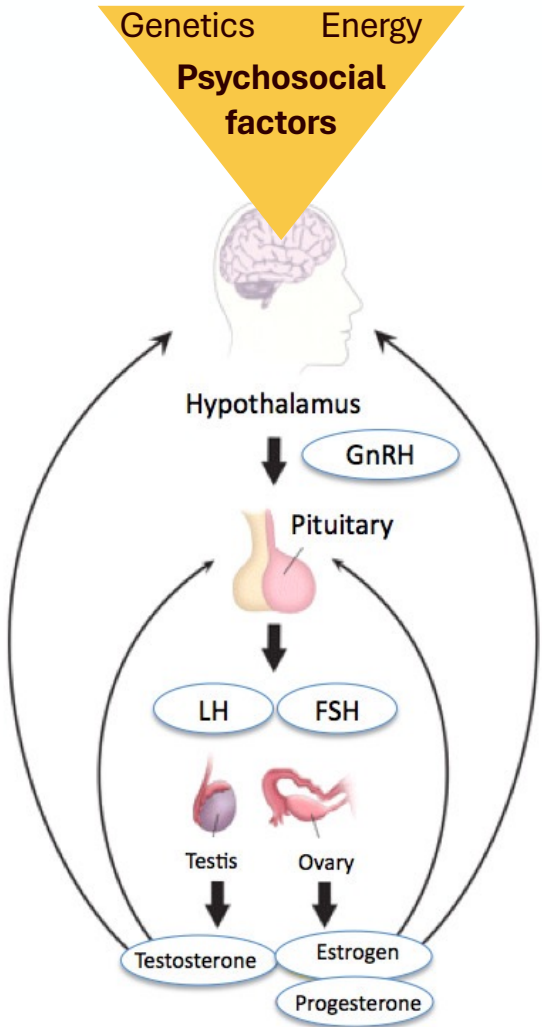
Role of psychosocial factors

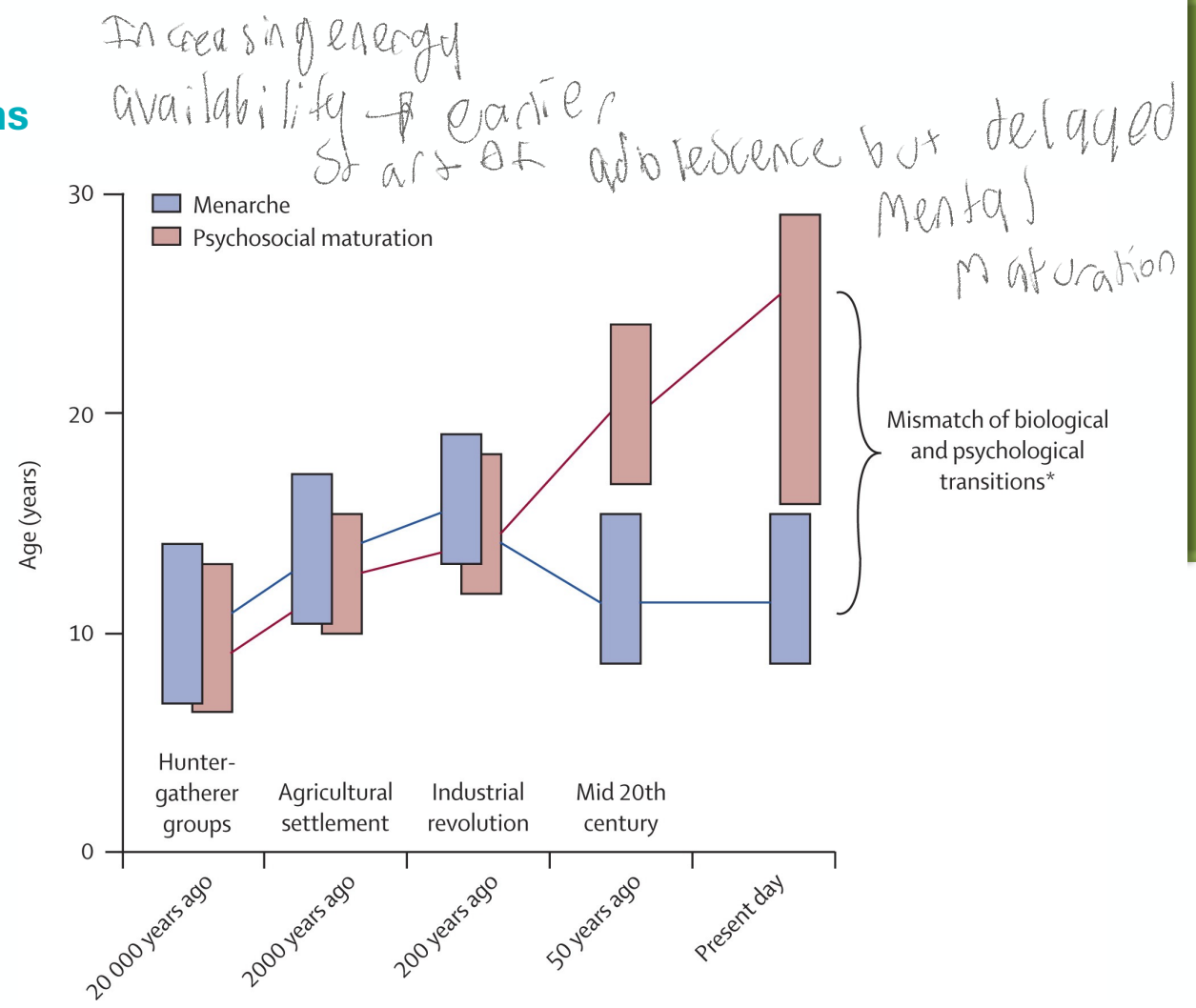
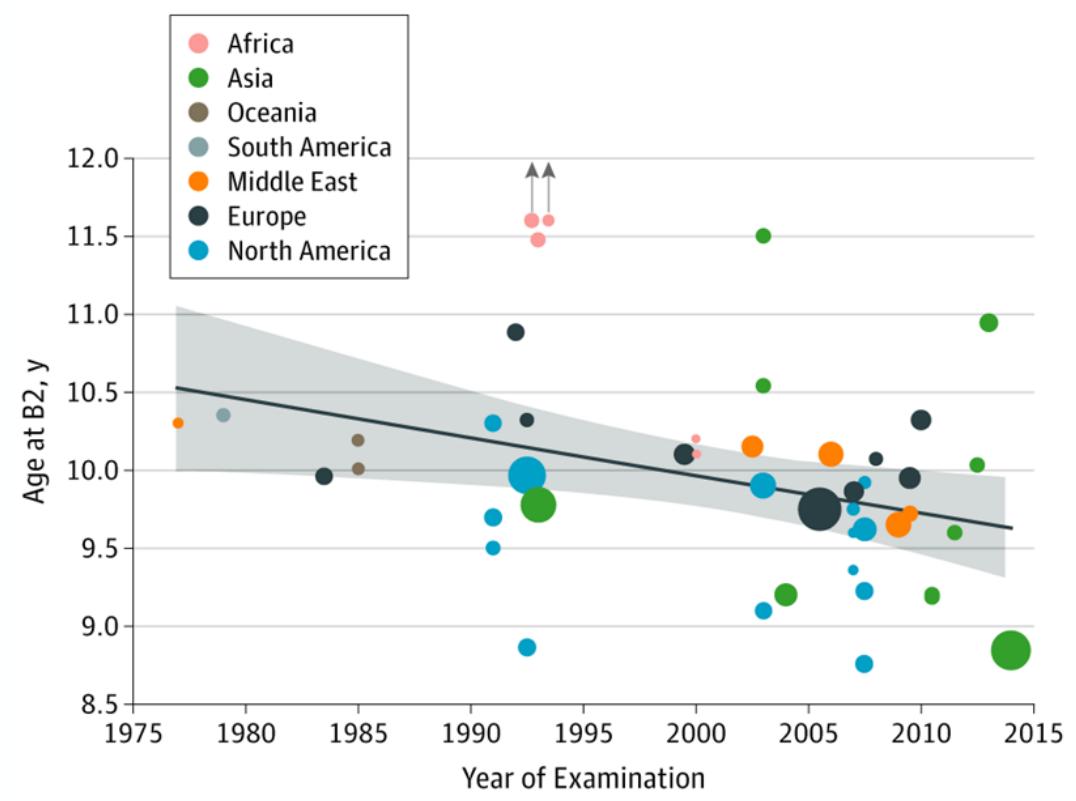
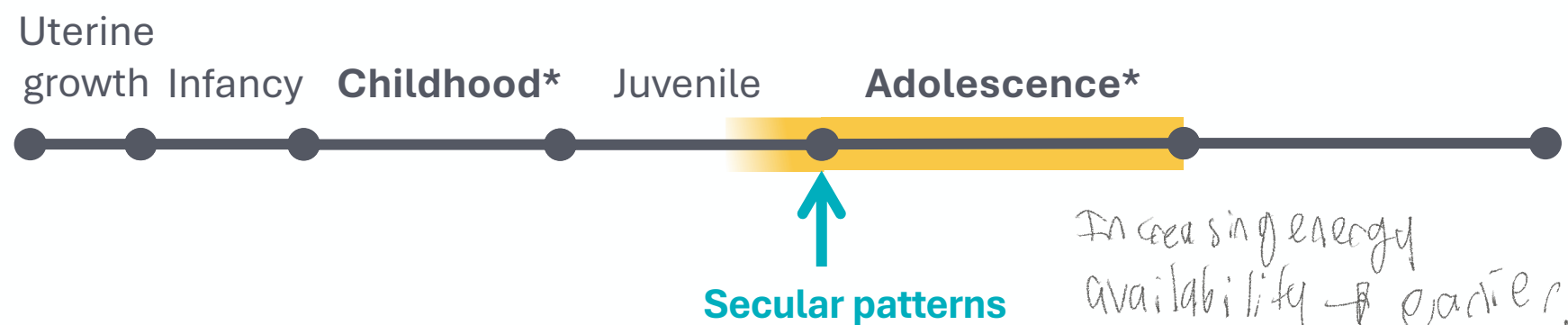
Found in one group, if society but not others!

Percentage of studies that found accelerating, delaying, mixed or no associations between **father absence** and the timing of puberty



Plus: endocrine pathways unclear



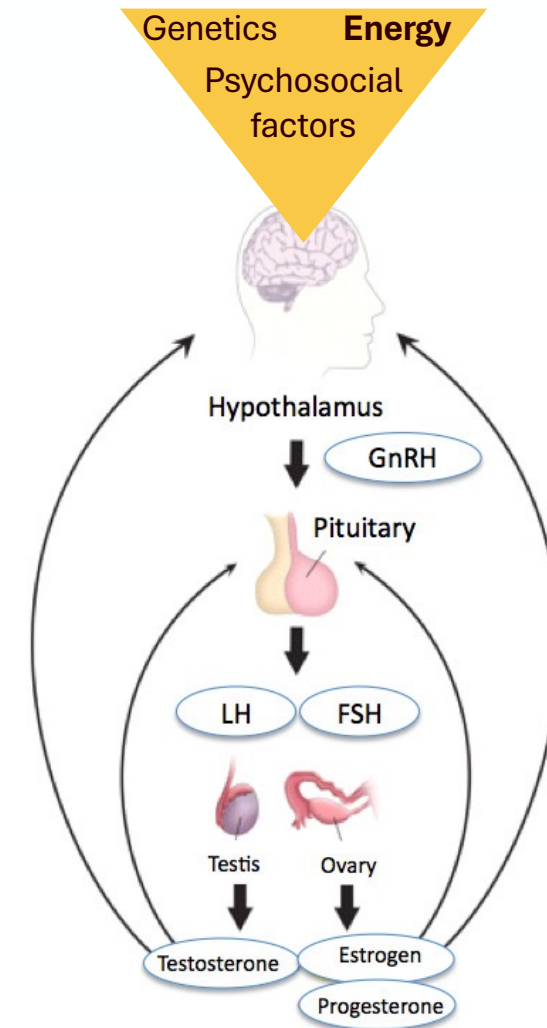


What is sex?

1. Chromosomal, gonadal, endocrine, genital, and phenotypic sex
2. There are strong correlations between these, but $R^2 \neq 1$!
3. There is no absolute binary at any of these levels apart from
 - the existence of the sex chromosomes (ish)
 - gametes

How does sex develop?

1. Chromosomal sex is set at fertilisation. The SRY gene on the Y-chromosome plays a key role in driving male differentiation
2. Followed by 6-week *indifferent period*. Some structures are bipotential (gonads and external genitalia); others aren't and degrade depending on SRY gene activation (Müllerian & Wolffian ducts)
3. 6+ weeks: sexual differentiation, driven largely by SRY – testes – T in males.
4. Minipuberty in infancy
5. Adolescence: divergent somatic & sexual growth patterns



What drives the timing of the **start** of adolescence?

5. **Adolescence**: divergent somatic & sexual growth patterns